☁️ Introduction to Amazon Web Services

AWS Technical Essentials — Module 1 of 7

📋 Module 1 Agenda

- Cloud computing fundamentals
- AWS global infrastructure
- Security and IAM essentials
- Lab: Introduction to IAM

☁️ Introduction to AWS Cloud

What is cloud computing and why does it matter?

🖥️ What Is Cloud Computing?

On-demand delivery of IT resources over the internet with **pay-as-you-go** pricing.

- No physical data centers to manage
- Provision resources in minutes
- Scale up or down on demand
- Pay only for what you consume

🎯 Six Advantages of Cloud

1. **Trade CapEx for variable expense**
2. **Massive economies of scale**
3. **Stop guessing capacity**
4. **Increase speed and agility**
5. **Stop running data centers**
6. **Go global in minutes**

🌍 AWS Global Infrastructure

🌏 Regions

- 33+ geographic locations
- Physically isolated
- Data stays in-Region

🏧 Availability Zones

- 105+ AZs globally
- One or more data centers
- Low-latency interconnects

⚡ Edge Locations

- 600+ Points of Presence
- CloudFront CDN caching
- Route 53 DNS hosting

🛠️ Interacting with AWS

💻 Console

- Web-based GUI
- Great for exploration

⌨️ CLI

- Command-line scripting
- Automate tasks

🔌 SDKs

- Language-specific APIs
- Integrate into apps

🔒 Security in the AWS Cloud

Shared responsibility and identity management

🔒 Shared Responsibility Model

AWS Responsibility

"Security OF the cloud"

- Physical data centers
- Hardware and networking
- Hypervisor and host OS

Your Responsibility

"Security IN the cloud"

- Data and encryption
- IAM and access control
- OS patches and firewalls

AWS Identity and Access Management

IAM controls *who* can access *what* in your AWS account.

- Global service (not Region-specific)
- Free to use, no additional charges
- Principle of least privilege
- Supports MFA for extra security

👥 Users, Roles, and Policies

👤 Users

- Represent a person or app
- Long-term credentials
- Console or programmatic

🎭 Roles

- Temporary credentials
- Assumed by services or users
- Preferred for EC2 apps

📜 Policies

- JSON permission documents
- Attach to users or roles
- Allow or deny actions

🛠️ Course Application

Employee Directory Application

Throughout this course, we build a web app layer by layer using AWS services.

- Module 2: EC2 for compute
- Module 3: VPC for networking
- Module 4: S3 for photo storage
- Module 5: DynamoDB for data

📋 Module 1 Summary

- Cloud = on-demand, pay-as-you-go IT
- AWS: Regions, AZs, Edge Locations
- Shared Responsibility Model
- IAM: users, roles, policies (least privilege)

🛠️ Lab Time

Introduction to AWS IAM

- Explore IAM users, groups, and roles
- Attach policies and test permissions
- Understand the effect of least privilege



Questions?

Module 1: Introduction to AWS